

Stony Brook University
Instructor: Alejandro Melo Ponce

Spring 2017

MIDTERM EXAM I

NAME: _____

ID: _____

Instructions: Answer all questions. The total number of points for the exam is 100. The number of points assigned to each section is indicated. Each question within a section carries the same weight. Calculators are allowed but other electronic devices or outside materials are NOT allowed. For questions with a numerical answer, give a numerical answer. For example

$$2 + V[X]$$

is NOT the final answer if you can compute a numerical value for $V[X]$.

You do not need to compute probabilities using the normal distribution. This is, your answer can be something like

$$1 - \Phi(1.67)$$

and you don't have to calculate that this number is actually 0.04746.

You are **NOT** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Part I: Multiple Choice Questions (20 points)

1. Analyzing the behavior of unemployment rates across U.S. states in March of 2006 is an example of using
 - a) time series data.
 - b) panel data.
 - c) cross-sectional data.
 - d) experimental data.
2. Ideal randomized controlled experiments in economics are
 - a) often performed in practice.
 - b) often used by the Federal Reserve to study the effects of monetary policy.
 - c) useful because they give a definition of a causal effect.
 - d) sometimes used by universities to determine who graduates in four years rather than five.
3. In econometrics, we typically do not rely on exact or finite sample distributions because
 - a) we have approximately an infinite number of observations (think of re-sampling).
 - b) variables typically are normally distributed.
 - c) the covariances of Y_i , Y_j are typically not zero.
 - d) asymptotic distributions can be counted on to provide good approximations to the exact sampling distribution (given the number of observations available in most cases).
4. The probability of an outcome
 - a) is the number of times that the outcome occurs in the long run.
 - b) equals $M \times N$, where M is the occurrences and N is the sample size.
 - c) is the proportion of times that the outcome occurs in the long run.
 - d) equals the sample mean divided by the sample standard deviation.
5. Two random variables X and Y are independently distributed if all of the following conditions hold, with the exception of
 - a) $\Pr(Y = y|X = x) = \Pr(Y = y)$.
 - b) knowing the value of one of the variables provides no information about the other.
 - c) if the conditional distribution of Y given X equals the marginal distribution of Y .
 - d) $E(Y) = E[E(Y)]$.
6. An estimator $\hat{\mu}$ of the population value μ is more efficient when compared to another estimator $\bar{\mu}$, if
 - a) $E[\hat{\mu}] > E[\bar{\mu}]$.
 - b) it has a smaller variance.
 - c) its c.d.f. is flatter than that of the other estimator.
 - d) both estimators are unbiased, and $V[\hat{\mu}] < V[\bar{\mu}]$.

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7. A type I error is
 - a) always less than one.
 - b) the error you make when rejecting the null hypothesis when it is true.
 - c) the error you make when rejecting the alternative hypothesis when it is true.
 - d) always 5%.
8. The t -statistic has the following distribution:
 - a) standard normal distribution for $n < 15$
 - b) Student t distribution when the sample is small.
 - c) Student t distribution if the Y is normally distributed and n is small.
 - d) a standard normal distribution if the sample standard deviation goes to zero.
9. The reason why estimators have a sampling distribution is that
 - a) economics is not a precise science
 - b) individuals respond differently to incentives
 - c) in real life you typically get to sample many times
 - d) the values of the explanatory variable and the error term differ across samples.
10. Finding a small value of the p -value (e.g. less than 5%)
 - a) indicates evidence in favor of the null hypothesis
 - b) implies that the t -statistic is less than 1.96.
 - c) indicates evidence against the null hypothesis
 - d) will only happen roughly one in twenty samples.

Part II: Open Problems (*80 points*)**Probability Distributions** (*20 points*)

Let X and Y be two random variables where X can be equal to 1 or 2 while Y can be equal to 1 or 4. Additionally, the joint probability distribution is given by $\Pr(X = 1, Y = 1) = 0.06$, $\Pr(X = 1, Y = 4) = 0.14$, and $\Pr(X = 2, Y = 4) = 0.56$.

(a) Find the marginal probability distributions of X and Y .

(b) Compute $E[X]$ and $V[X]$.

(c) Compute $E[X|Y = 4]$.

(d) Compute $\text{Cov}(X, Y)$.

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Functions of Random Variables and Estimators (30 points)

The random variable W has a mean μ and a variance σ^2 . You randomly draw two observations from the distribution of W , namely W_1 and W_2 . Consider the estimator of μ given by $\hat{\mu} = 1.5W_1 - 0.5W_2$

- Compute the mean $E[\hat{\mu}]$ and variance $V[\hat{\mu}]$ in terms of μ and σ^2 .
- Is $\hat{\mu}$ an unbiased estimator of μ ? Why?
- Is $\hat{\mu}$ a more efficient estimator than the sample average $\overline{W}$? Hint: What is the variance of $\overline{W}$? If you do not remember it you can still answer this question.

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Large Samples and Hypothesis Tests (30 points)

- a) Explain the central limit theorem (CLT) briefly and in your own words (just writing down a formula will yield no points). Why is this result so important?
- b) You want to learn about μ_Y for which you draw a random sample $Y_1, \dots, Y_{100}$. With the sample information you computed $\bar{Y}^{act} = 50$ and $s^2 = 25$. Compute the p -value for the test $\mu_Y = \mu_{Y,0} = 51$ against the alternative $\mu_Y = \mu_{Y,0} \neq 51$.
- c) Suppose you find that $p\text{-value} = 0.03$ for a two-sided hypothesis test on the population mean. For which significance levels α you fail to reject the null hypothesis?
- d) Calculate the 90% confidence interval for the population mean. How would you interpret this confidence interval? How can you use it to test a hypothesis about the mean?

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Formulas

$$E[Y] = \mu_Y = \sum_{i=1}^K p_i y_i \quad (1)$$

$$V[Y] = \sigma_Y^2 = E[(Y - \mu_Y)^2] = \sum_{i=1}^K p_i (Y_i - \mu_Y)^2 \quad (2)$$

$$\Pr(Y = y|X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} \quad (3)$$

$$E[Y|X = x] = \sum_{i=1}^K \Pr(Y = y_i|X = x) y_i \quad (4)$$

$$V[Y|X = x] = \sum_{i=1}^K \Pr(Y = y_i|X = x) [y_i - E[Y|X = x]]^2 \quad (5)$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] \quad (6)$$

$$E[a + bX + cY] = a + b\mu_X + c\mu_Y \quad (7)$$

$$V[a + bY] = b^2\sigma_Y^2 \quad (8)$$

$$V[aX \pm bY] = a^2\sigma_X^2 + b^2\sigma_Y^2 \pm 2ab\text{Cov}(X, Y) \quad (9)$$

$$E[Y^2] = \sigma_Y^2 + \mu_Y^2 \quad (10)$$

$$\text{Cov}[a + bX + cY, Y] = b\sigma_{XY} + c\sigma_{YY} \quad (11)$$

$$E[XY] = \sigma_{XY} + \mu_X\mu_Y \quad (12)$$

$$p\text{-value} = \Pr_{H_0}(|\bar{Y} - \mu_{Y,0}| > |\bar{Y}^{act} - \mu_{Y,0}|) \quad \text{two-sided} \quad (13)$$

$$p\text{-value} = \Pr_{H_0}((\bar{Y} - \mu_{Y,0}) > (\bar{Y}^{act} - \mu_{Y,0})) \quad \text{one-sided} \quad (14)$$

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (15)$$

$$\sigma_Y^2 = \frac{\sigma_Y^2}{n} \quad (16)$$

$$SE(\bar{Y}) = \bar{s} = \frac{s_Y}{\sqrt{n}} \quad (17)$$

$$z = \frac{\bar{Y} - \mu_{Y,0}}{\sigma_Y} \quad (18)$$

$$t = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})} \quad (19)$$

$$90\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.64 \times SE(\bar{Y})] \quad (20)$$